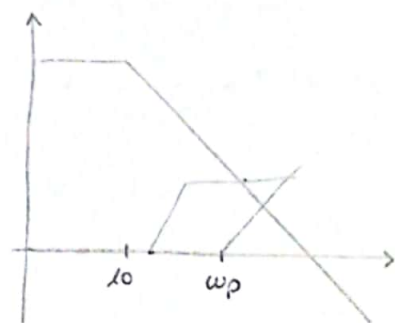


② $A_v(s) = -\frac{R_2}{\frac{1}{sC_1}} = -sC_1 R_2$

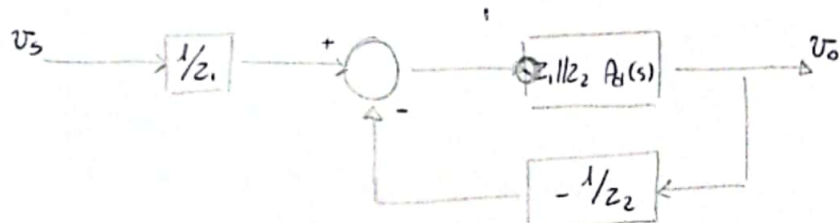


$T = -\frac{x_r}{x_s} = A_d(s) \frac{Z_1}{Z_1 + Z_2} = A_d(s) \frac{\frac{1}{sC_1}}{\frac{1}{sC_1} + R_2} = A_d(s) \frac{1}{1 + sC_1 R_2}$

$\omega_p = \frac{1}{C_1 R_2} = 454.55 \text{ rad/s}$

$\left\{ \begin{array}{l} A_{do} \omega_0 = \frac{1}{\beta} \omega_{cr} \\ \frac{1}{\beta} \frac{1}{\omega_{cr}} = \frac{1}{\omega_p} \end{array} \right. \rightarrow \omega_{cr} = \sqrt{A_{do} \omega_0 \omega_p} = 21.32 \text{ rad/s}$

$PH = 180 - \text{atan} \frac{\omega_{cr}}{\omega_0} - \text{atan} \frac{\omega_{cr}}{\omega_p} = 1.25^\circ$



$Z_1 = \frac{1}{sC_1} \rightarrow R_2(s) = \frac{1}{Z_1} = sC_1$

$\tan 60^\circ = 1.73 \rightarrow \omega_z = \frac{\omega_{cr}}{2}$

$\beta_{\infty} = \frac{R_1}{R_1 + R_2} \rightarrow \frac{1}{\beta_{\infty}} = 1 + \frac{R_2}{R_1}$

$T' = A_d(s) \cdot \frac{R_1 + \frac{1}{sC_1}}{R_1 + R_2 + \frac{1}{sC_1}} = A_d(s) \frac{1 + sC_1 R_1}{1 + sC_1 (R_1 + R_2)}$

$A_{do} \omega_0 = \frac{1}{\beta_{\infty}^2} \omega_{cr}^2 = \frac{1}{\beta_{\infty}^2} 2\omega_z \rightarrow A_{do} \omega_0 = \frac{R_1 + R_2}{R_1} 2 \frac{1}{C_1 R_1} \rightarrow A_{do} \omega_0 = \frac{(R_1 + R_2) 2}{C_1 R_1^2}$

$\frac{1}{\omega_p} = \frac{1}{\beta_{\infty}^2} \frac{1}{\omega_z} = \frac{1}{\beta_{\infty}^2} \frac{1}{\omega_{cr}} = \frac{1}{\beta_{\infty}^2} \frac{2}{2\omega_z}$

$\frac{C_1 R_1^2}{A_{do} \omega_0} - \frac{2R_1}{A_{do} \omega_0} - \frac{2R_2}{A_{do} \omega_0} = 0 \rightarrow R_1 =$

$\left\{ \begin{array}{l} A_{do} \omega_0 = A_x 2\omega_z' \\ \frac{1}{\omega_p'} = A_x \frac{1}{\omega_z'} \end{array} \right. \rightarrow A_x = \frac{\omega_z'}{\omega_p'} \rightarrow A_{do} \omega_0 = 2 \frac{\omega_z'^2}{\omega_p'} = \frac{2 \left(\frac{1}{C_1 R_1} \right)^2}{\frac{1}{C_1 (R_1 + R_2)}} = \frac{2 C_1^2 (R_1 + R_2)}{C_1^2 R_1^2}$

$\rightarrow A_{do} \omega_0 R_1^2 = \frac{2 [R_1 + R_2]}{C_1} \rightarrow R_1^2 C_1 A_{do} \omega_0 - 2R_1 - 2R_2 = 0 \rightarrow R_1 = \begin{cases} 673.4 \Omega \\ -653.4 \end{cases}$

$\omega_z = \frac{1}{R_1 C_1} = 14.85 \text{ rad/s} \quad \omega_{cr} = 29.7 \text{ rad/s} \quad \omega_p = \frac{1}{C_1 [R_1 + R_2]} = 441.05 \text{ rad/s}$

$\frac{1}{\beta_{\infty}} = 33.67 \quad \omega_{cr}^c = A_{do} \omega_0 \beta_{\infty} = 29.7 \text{ rad/s}$

$PH = 180 - \text{atan} \frac{\omega_{cr}^c}{\omega_0} - \text{atan} \frac{\omega_{cr}^c}{\omega_p} + \text{atan} \frac{\omega_{cr}^c}{\omega_z} = 64.3^\circ$

1 ora 15 min